

Full public-denominator external comparison. CIFAR-GIST-512, 60K self-join, pageable-host FP32 input through sorted canonical host IDs; 3,926,078 exact directed pairs.

Method	Arithmetic / output	p10 (s)	Median (s)	p90 (s)	Exact	Relative to TensorJoin
TensorJoin	certified mixed precision	0.924369	0.926153	0.928841	yes	1.000x
MiSTIC	FP64	4.631841	4.732177	4.770933	yes	5.083x slower [5.030, 5.133]
GDS-Join	FP64	10.820870	10.927797	10.990203	yes	11.783x slower [11.716, 11.846]
FaSTED	FP16 / FP32; changed output	0.200003	0.229410	0.231410	no	4.037x faster (context only)

FaSTED has 598 false negatives and 1,130 false positives. Exact-system intervals are paired 95% round-bootstrap intervals; FaSTED is approximate context only.

Table 1. The scalable G2B implementation outperforms both exact external systems under the complete public denominator, whereas the faster FaSTED point changes the output set.